Capstone Phase–II Roadmap

🎯 Phase–II Scope

What We Will Do in Phase–II

- Complete high-level architecture and system design
- Fully implement the Gmail module
- Implement the Chrome History module up to a partial to intermediate level

What Is Moved to Phase–III

- YouTube integration
- Complete Chrome History enhancement
- Complete digital well-being module

🗓️ Phase–II Weekly Plan (Up to Review-1)

Review-1 Date: 16 February 2026

Week 1 – Architectural Thinking & System Understanding

🗓️ **20 January – 26 January 2026**

Goal: Build a clear mental model of how the entire system works.

What We Will Do:

- Understand the system end-to-end
- Break the system into major parts:
  - Data ingestion
  - Processing
  - Storage
  - Search and retrieval
- Decide responsibilities of each part
- Think through data flow from start to end
- Discuss edge cases (permissions, missing data, failures)
- Write a conceptual explanation of the system (no diagrams yet)

Week 2 – Architecture & High-Level Design

🗓️ **27 January – 2 February 2026**

Goal: Convert the system understanding into a formal design.

What We Will Do:

- Create system-level architecture diagrams
- Create data-flow diagrams
- Define how Gmail and Chrome History data flow through the system
- Define preprocessing, storage, and retrieval flow
- Define Chrome extension interaction
- Finalize privacy and boundary decisions
- Complete the High-Level Design (HLD) document

Week 3 – Gmail Module: Design & Initial Implementation

🗓️ **3 February – 9 February 2026**

Goal: Start building the system using Gmail as the reference module.

What We Will Do:

- Study Gmail API authentication and permissions
- Decide what Gmail data to use
- Design Gmail module at a low level
- Set up project repository and structure
- Start Gmail implementation:
  - OAuth setup
  - Email metadata ingestion
  - Basic preprocessing

Week 4 – Gmail Progress & Review-1 Preparation

🗓️ **10 February – 15 February 2026**

Goal: Stabilize Gmail work and prepare for Review-1 and ISA-1.

What We Will Do:

- Improve Gmail ingestion reliability
- Clean up data models and code
- Avoid adding risky new features
- Prepare Review-1 explanation:
  - Problem statement
  - Architecture overview
  - Phase–II objective
- Ensure HLD matches implementation

- Internal discussion and Q&A practice

🏁 Expected State at Review-1 (16 February 2026)

- High-level architecture and design completed
- HLD finalized
- Gmail module started and visibly progressing
- Clear plan for remaining Phase–II work

🚀 Phase–II Weekly Plan (After Review-1)

Working Window: 9 March – 23 April 2026

(Work starts after ISA-1; final ESA on 23 April)

Week 5 – Gmail Completion

🗓️ **9 March – 15 March 2026**

Goal: Complete the Gmail module fully.

What We Will Do:

- Complete Gmail ingestion flow
- Finalize preprocessing logic
- Ensure correct storage and retrieval
- Clean Gmail data models
- Validate end-to-end Gmail flow
- Fix known issues
- Align Gmail implementation with HLD

Week 6 – Chrome History: Understanding & Initial Implementation

🗓️ **16 March – 22 March 2026**

Goal: Understand Chrome History data and begin implementation.

What We Will Do:

- Study Chrome History API behavior
- Decide what browsing data to capture
- Design Chrome History low-level structure
- Start Chrome History ingestion
- Store basic browsing metadata

Week 7 – Chrome History: Core Integration

🗓️ **23 March – 29 March 2026**

Goal: Integrate Chrome History into the system pipeline.

What We Will Do:

- Improve ingestion reliability
- Implement preprocessing and normalization
- Connect Chrome History data to storage
- Enable basic retrieval
- Ensure architecture alignment

Week 8 – Chrome History: Intermediate-Level Search

🗓️ **30 March – 5 April 2026**

Goal: Bring Chrome History to a solid intermediate level.

What We Will Do:

- Improve search quality
- Handle noisy and repetitive data
- Improve relevance of results
- Test unified behavior with Gmail
- Stabilize Chrome History module

Week 9 – Chrome History Refinement & System Stability

🗓️ **6 April – 12 April 2026**

Goal: Refine and stabilize the Phase–II system.

What We Will Do:

- Bug fixes and performance improvements
- Handle edge cases
- Clean codebase and data models
- Prepare internal demo flows

Week 10 – Final ESA Review Preparation

🗓️ **13 April – 23 April 2026**

Goal: Prepare confidently for the Final ESA Review.

What We Will Do:

- Final testing of Gmail and Chrome History
- Minor fixes only (no new features)

- Documentation updates
- Prepare ESA explanations:
  - Problem statement
  - Architecture overview
  - Phase–II outcomes
- Dry runs and internal Q&A

🏆 Final State by 23 April 2026 (ESA Review)

- Gmail fully implemented
- Chrome History at a solid intermediate level
- Unified search working across Gmail and Chrome
- Architecture validated through implementation
- Ready to move into Phase–III